A Multiscale Spectral Features Graph Fusion Method for Hyperspectral Band Selection

Weiwei Sun, *Senior Member, IEEE*, Gang Yang, Jiangtao Peng, *Senior Member, IEEE*, Xiangchao Meng, *Member, IEEE*, Ke He, Wei Li, *Senior Member, IEEE*, Heng-Chao Li, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— This article proposes a multiscale spectral features graph fusion (MSFGF) method for selecting proper hyperspectral bands. The MSFGF regards that the selected bands should reflect diagnostic spectral information of ground objects at different scales, and it explores band selection from the aspect of multiple spatial scales. First, it adopts the multiscale low-rank decomposition (MSLRD) model to find multiscale spectral features of different ground objects. The model considers divergent spatial structures or spatial correlations of ground objects at different scales, and factorizes the hyperspectral data cube into a series of low-rank block-wise data cubes, where the blocks take spatial structures of different ground objects at increasing scales. Second, the MSFGF presents the multiscale sparse spectral clustering (MSSC) model to fuse the separate connected graphs of multiscale spectral features into a consensus graph. The consensus graph combines the complementary information of multiscale spectral features and helps to reveal the intrinsic clustering structure of all spectral bands. Finally, the MSFGF utilizes spectral clustering to find clusters from the consensus graph and selects representative bands. Experimental results on three widely used hyperspectral data prove the superiority of MSFGF in selecting bands, where it outperforms other seven state-of-the-art methods in classification with an acceptable computational cost.**

***Index Terms*— Band selection, hyperspectral imagery (HSI), multiscale low-rank decomposition (MSLRD), multiscale sparse spectral clustering (MSSC), multiscale spectral features graph fusion (MSFGF).**

Manuscript received February 5, 2021; revised June 18, 2021 and August 1, 2021; accepted August 1, 2021. Date of publication August 13, 2021; date of current version January 26, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41971296, Grant 41801256, and Grant 61871177 and in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR19D010001 and Grant LQ18D010001. *(Corresponding authors: Gang Yang, Jiangtao Peng.)*

Weiwei Sun, Gang Yang, and Ke He are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo, Zhejiang 315211, China (e-mail: nbsww@outlook.com; yanggang@nbu.edu.cn; hk2098@yeah.net).

Jiangtao Peng is with Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Xiangchao Meng is with the Faculty of Electrical Engineering and Computer Science, Ningbo 315211, China (e-mail: megnxiangchao@nbu.edu.cn).

Wei Li is with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org).

Heng-Chao Li is with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 611756, China (e-mail: lihengchao_78@163.com).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2021.3102246

I. Introduction

HYPERSPECTRAL imaging, as an advanced on-board or airborne remote sensing technique, can capture spectral information of ground objects with hundreds of narrow bands [1], [2]. The collected hyperspectral imagery (HSI) can be used in many practical applications, e.g., precise agriculture [3], urban mapping [4], geological exploration [5], and military surveillance [6]. However, the high intra-correlation bands render that the HSI data suffers from the severe problem of "curse of dimensionality," where the classification performance may decrease with the continuously increasing dimensionality of spectral bands [7], [8]. Accordingly, dimensionality reduction using feature extraction or band selection is a good choice for conquering the above problem [9].

Band selection selects proper representative bands from the original band set, and the dimensionality-reduced data has clearer spectral meanings than that of feature selection [10], [11]. Therefore, we focus our topic on band selection. Remote sensing literature has reported many band selection methods, which can be roughly grouped into six categories: ranking-based, clustering-based, searching-based, embedding learning-based, sparsity-based, and hybrid-based methods [12]. Ranking-based methods quantify the significance of each band and choose the top-ranked ones, e.g., maximum-variance principal component analysis (MVPCA) and manifold ranking-based band selection algorithm [13]. The selected bands highly correlate with the criterion that is used to rank the bands. Clustering-based methods select highly uncorrelated bands from each band cluster, and their typical examples are the fast density-peak-based clustering-based method (FDPC) [14] and the fast neighborhood grouping method [15]. The selected bands can avoid high correlations but do not guarantee maximal spectral information, and they highly depend on the clustering techniques. Searching-based methods formulate band selection into an optimization program to search a proper band subset. The newly presented searching-based algorithms are metaheuristic optimization [16] and Maximum-Information-Minimum-Redundancy with modified Discrete Gravitational Search Algorithm [17]. The searching program always brings about high computational complexity. Embedding learning-based methods embed the band selection term into the optimiza-

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

tion program of specific application models (e.g., classifier and target detection), and simultaneously select bands while obtaining an optimal application model. Popular algorithms include recursive support vector machines [18] and weighted kernel regularization [19]. Recently, deep learning-based methods have been used in many fields and obtained good performance for hyperspectral band selection. These methods learn complex underlying relationships between bands, and then the band subset is obtained by network mapping. Typical examples are band-wise-independent convolution and hard thresholding convolutional neural network (BHCNN) [20] and End-to-End BS-Net [21]. However, these methods usually need a large number of training samples and have low computational efficiency. Sparsity-based methods regard that each band can be sparsely represented by several other bands, and the estimated sparse coefficients reveal certain underlying structures within the HSI bands and are used for band selection. Typical methods include sparse non-negative matrix factorization with thresholded earth's mover distance (SNMF-TEMD) [22], fast and robust self-representation [23], and non-negative low-rank representation (NNLRR) [24]. The sparse coefficients are sensitive to the convergence of defined optimization programs. Hybrid-based methods combine multiple schemes to select appropriate bands, e.g., combining clustering with ranking schemes like intra-class separability index (SSI) [25] and density-based spatial clustering of applications with noise (DBSCAN) [26], combing ranking with searching schemes like conditional mutual information and genetic algorithm [27] and minimum redundancy maximum relevance [28].

In recent years, graph learning methods have been introduced into the hyperspectral field [9], [29], [30], which explore the cluster structures of HSI data that lie in some low-dimensional subspaces to the select bands. In these methods, a graph encoded by an affinity matrix is usually constructed to reveal band correlation, and then spectral clustering is performed to find the clusters and choose representative bands [31]. The performance of graph learning methods is largely determined by the quality of the constructed graph or the affinity matrix. In [32], the HSI data was assumed to lie in a unified low-dimensional subspace, and low-rank representation was estimated to learn the graph. Later, the improved sparse subspace clustering (ISSC) [33] regarded that all bands are sampled from some low-rank independent subspaces and implemented the self-representation model to depict the intra-band correlations. Each band can be represented by a combination of other bands in its own subspace, and the sparse coefficient vectors are used to construct an improved affinity matrix. After that, the spatial-spectral information between pair-wise pixels was added to weigh the entries in the sparse coefficients. The formulated squared weighted low-rank subspace clustering (SWLRSC) band selection [34] algorithm improved from ISSC, and adopted the squared weighted strategy to construct a strongly connected adjacent matrix with sparse coefficients. More recently, the fast and latent low-rank subspace clustering (FLLRSC) [11] considers noisy bands and insufficient samples of underlying samples in ISSC, and introduces hidden or unobserved bands to construct the latent low-rank subspaces for promoting the performance of the correntropy similarity matrix. The above low-rank methods formulate the HSI data into a 2-D real matrix and utilize intra-band correlations or band similarity of the whole spectral band set.

All these methods have made significant progress in exploiting the clustering structure of HSI bands using the learned graph, but they still have some drawbacks. As we all know, ground objects have different spatial sizes and divergent diagnostic spectral bands along with the changing wavelengths from visible to short-wave infrared, which may affect the selection of dissimilar and representative spectral bands. For example, both a small-size building and widely scattered vegetations equally contribute to the mapping of coastal wetlands. Accordingly, the selected bands from graph learning methods should reflect the diagnostic spectral features from both classes. Unfortunately, the above graph learning methods did not involve the spatial-scale issue of various ground objects in the image scene when constructing the connected graph with spectral bands. Their constructed graph only demonstrates the clustering structure of spectral information of all ground objects at a unified spatial scale, which may ignore diagnostic bands for specific ground objects with various spatial scales, and then degrade the classification behaviors of selected bands.

Therefore, in this article, we propose a multiscale spectral features graph fusion (MSFGF) for solving the above problem of hyperspectral band selection. The idea of MSFGF is to find the low-rank spectral features of ground objects at various spatial scales, and to fuse their separate connected graphs into a consensus graph and choose proper bands using spectral clustering on the fused graph. First, the MSFGF adopts the multiscale low-rank decomposition (MSLRD) model to find multiscale spectral features of different ground objects. The model considers divergent spatial structures or spatial correlations of ground objects at different scales and factorizes the HSI data cube into a series of low-rank block-wise data cube, where the low-rank blocks take spatial structures of different ground objects at increasing scales. After that, the model implements multiscale sparse spectral clustering (MSSC) to fuse the connected graphs of multiscale spectral features into a unified connected graph. The fused graph combines the clustering structures of all spectral features with different spatial scales, and reveals the real clustering structure of all spectral bands. Finally, spectral clustering is adopted to find clusters from the fused graph and the representative bands are selected. Compared with all previous studies, our proposed MSFGF favors three main contributions.

1) Our MSFGF is the first to carefully consider the spatial-scale issue of ground objects in selecting HSI bands. It explores various clustering structures of spectral features at different spatial scales, and combines them to obtain the fused graph to promote the behaviors of band selection.
2) Our MSFGF adopts the MSLRD model to extract various spectral features at different spatial scales. The MSLRD implements a block reshape operator to extract spatial structures at different scales, and decomposes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

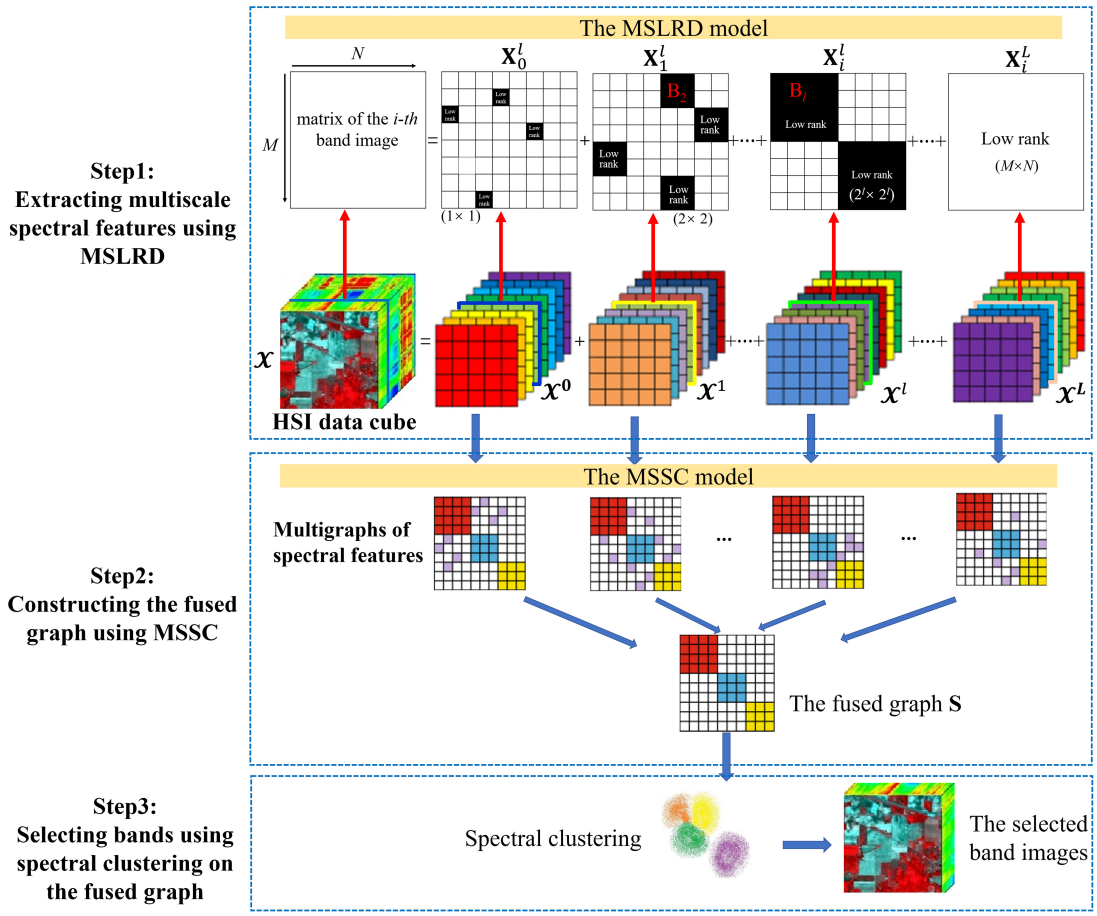


Fig. 1. Flowchart of MSFGF for band selection.

the HSI data into a series of low-rank block-wise data cubes. The obtained feature data cubes take spectral feature information of ground objects at increasing scales.

3) Our MSFGF implements MSSC to fuse the connected graphs of multiscale spectral features into a consensus graph. Inspired from the ISSC method, the MSSC not only considers different approximation levels and contributions from each connected graph to the fused one, but also guarantees the clustering property of all spectral bands on the desired consensus graph.

The rest of this article is arranged as follows. Section II describes the MSFGF for hyperspectral band selection. Section III presents experimental results on three hyperspectral datasets. Section IV draws conclusions from this article.

TABLE I
NOTATIONS AND DEFINITIONS

Notation	Definition
$\boldsymbol{\mathcal{X}}$	HSI data
$\mathbf{Y}_i$	i-th band image of $\boldsymbol{\mathcal{X}}$
$\mathbf{M}$	Selected bands
$\mathbf{R}_b(\cdot)$	Block reshape operator
$\mathbf{B}_l$	Low-rank block matrices
$\mathbf{R}_b^{\mathrm{T}}(\cdot)$	Block embedding operator
$\{\mathbf{Y}^l\}_{l=0}^L$	The estimated multiscale low-rank matrices from $\boldsymbol{\mathcal{X}}$
$\mathbf{Z}^l$	The connected graph matrix
$\mathbf{S}$	The fused graph
$\mathbf{L}$	Laplacian matrix
$\mathrm{Tr}(\cdot)$	Trace operator
$\alpha, \beta,$ and γ	Regularization parameters

II. Multiscale Spectral Features Graph Fusion

Consider the HSI as a data cube denoted as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{M\times N\times D}$, where $\mathbf{Y}_i$ is the ith band, M and N are the length and width of a single band image, respectively, and D is the number of bands. Band selection is to select k proper bands (i.e., band images) $\mathbf{M} = \boldsymbol{\mathcal{X}}(:,:,\theta) \in \mathbb{R}^{M\times N\times k}$ from the original data cube, where $\theta \in \{1, 2, \ldots, D\}$ and $|\theta| = k$. Table I shows the notations in this article. The idea of MSFGF is to extract spectral features of ground objects at different spatial scales, fuse the connected graphs of multiscale spectral features into a consensus graph, and make spectral clustering on the fused graph to select proper bands. Fig. 1 illustrates the flowchart of our MSFGF.

A. Extracting Multiscale Spectral Features Using MSLRD

Ground objects with different spatial scales have their specific diagnostic spectral features [35], [36], that should be reflected by the representative bands. Multiscale analysis can help us to understand the spatial structures of each ground

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

objects accurately and compactly at different scales. Therefore, we adopt the MSLRD model [37] to extract spectral features with different spatial scales. The MSLRD decomposes each band image into a series of low-rank block-wise matrices, where the low-rank blocks depict spatial structures of different ground objects at increasing scales.

For the full data matrix $\mathbf{X_i} \in \mathbb{R}^{M\times N}$ of the ith band image, the MSLRD implements a block reshape operator $\mathbf{R_b}(\cdot)$ [38] to extract $L+1$ different low-rank block matrices $\{\mathbf{B^l}\}_{l=1}^{L}$ from $\mathbf{X}$, and then embeds the blocks into the full zero matrix to formulate the partitions $\{\mathbf{X}_i^l\}_{l=1}^{L}$ at different spatial scales. Each low-rank block $\mathbf{B}_l$ has an order-magnitude larger than those in its previous scale and has the size of $m_l \times n_l$. For the simplicity, we set $m_l = n_l$ and set m_l as the powers of 2, e.g., $m_l = 2^l$, and the largest block size is $M \times N$. Accordingly, each band image $\mathbf{X_i}$ can be decomposed into a series of data matrices $\{\mathbf{X}_i^l\}_{l=0}^{L} \in \mathbb{R}^{M\times N}$, in which each $\mathbf{X}_i^l$ is block-wise low-rank with respect to its partition [39], i.e.,

$$\mathbf{X_i} = \sum_{l=0}^{L} \mathbf{X}_i^l, \text{ s.t., } \mathbf{X}_i^l = \sum \mathbf{R}_b^{\mathrm{T}}(\mathbf{B}_l), \mathbf{B}_l = \mathbf{U}_b\mathbf{S}_b\mathbf{V}_b^{\mathrm{T}} \tag{1}$$

where $\mathbf{U}_b \in \mathbb{R}^{m_l\times r_b}$, $\mathbf{S}_b \in \mathbb{R}^{r_b\times r_b}$, and $\mathbf{V}_b \in \mathbb{R}^{r_b\times n_l}$ are resulted from the SVD decomposition of $\mathbf{B}_l$, which guarantees the low-rank property as r_b, and $\mathbf{R}_b^{\mathrm{T}}(\cdot)$ is the block embedding operator. Each partition matrix $\mathbf{X}_i^l$ illustrates detailed spatial structures of ground objects at the lth scale. Accordingly, if we formulate the partitioned data cube $[\mathbf{X}_i^l]_{i=1}^{D} \in \mathbb{R}^{M\times N\times D}$ into the 2-D matrix, all the bands can be regarded to take spectral features of ground objects at the lth spatial scale accordingly.

Considering the good approximation of nuclear norm to the rank optimization [40], (1) can be optimized by minimizing the block-wise nuclear norm of each partitioned matrix $\mathbf{X}_i^l$, i.e.,

$$\min_{\mathbf{x}_i^0,\mathbf{x}_i^1,\ldots,\mathbf{x}_L^0} \sum_{l=0}^{L} \alpha_l \sum_{l=0}^{L} \left\|\mathbf{R_b}(\mathbf{X}_i^l)\right\|_*, \text{ s.t., } \mathbf{X_i} = \sum_{l=0}^{L} \mathbf{X}_i^l \tag{2}$$

where $\sum_{l=0}^{L} \|\mathbf{R_b}(\mathbf{X}_i^l)\|_*$ is the block-wise nuclear norm of $\mathbf{X}_i^l$ for the lth scale, $\|\cdot\|_*$ is the nuclear norm operator, and λ_l is the regularization parameter. The α_l has been proven to have the Gaussian complexity of each norm $\|\cdot\|_{(*)}$, which can be the maximum singular value of a random Gaussian matrix. Accordingly, we follow previous studies by Bandeira and Handel [41] and set the regularization parameter $\alpha_l \sim (m)_i^{1/2} + (n)_i^{1/2} + (\log(MN/\max)(m_i, n_i))^{1/2}$. The first partitioned matrix $\mathbf{X}_i^0$ has the 1×1 block size with $\alpha_0 = (\log)^{1/2}(MN)$, and the largest partitioned matrix $\mathbf{X}_i^L$ has the $M \times N$ block size with $\alpha_L \sim (M)^{1/2} + (N)^{1/2}$. The problem in (2) is convex, and can be solved by the alternating direction method of multipliers (ADMM) [42]. More details about the solution can be found in our previous paper [37]. The estimated multiscale low-rank matrices $\{\mathbf{Y}^l\}_{l=0}^{L} \in \mathbb{R}^{MN\times D}$ from $\boldsymbol{\mathcal{X}}$ reflects the spectral features of ground objects at diverse spatial scales, where $\mathbf{Y}^l$ is the matrix format of partitioned data cube $\boldsymbol{\mathcal{X}}^l = [\mathbf{X}_i^l]_{i=1}^{D} \in \mathbb{R}^{M\times N\times D}$ at the lth spatial scale.

B. Constructing the Fused Consensus Graph Using MSSC

The multiscale low-rank matrices $\{\mathbf{Y}^l\}_{l=0}^{L}$ depict spectral features of ground objects at different spatial scales. Their constructed connected graphs by the affinity matrix reflect the clustering of spectral features at different spatial scales and contribute to revealing the real inner relationship between all bands. Therefore, in this article, inspired by the ISSC [33] method and multiview learning [43], we present the MSSC method to construct the fused graph $\mathbf{S}$ from multiscale spectral features to combine the complementary information of spectral features $\mathbf{Y}^l$ at different spatial scales. The MSSC makes use of the self-representation of $\mathbf{Y}^l$, and considers both different approximation levels and contributions of each connected graph $\mathbf{Z}^l$ to the fused graph $\mathbf{S}$, to uncover the real clustering structure of all spectral bands.

For each spectral feature matrix $\mathbf{Y}^l$, we regard that its each-column feature band (i.e., vector) $\mathbf{y}^l$ can be sparsely expressed as a linear combination of other bands in the same low-dimensional subspace. The sparse coefficient indicates the similarities between spectral bands at different spatial scales. Accordingly, inspired by ISSC, the coefficient or graph matrix $\mathbf{Z}^l$ can be obtained by solving the following problem [44]:

$$\min_{\mathbf{Z}^l} \sum_{l=0}^{L} \left\|\mathbf{Y}^l - \mathbf{Y}^l\mathbf{Z}^l\right\|_F^2 + \alpha\left\|\mathbf{Z}^l\right\|_F^2, \quad \text{s.t., } \mathbf{Z}^l \geq 0, \operatorname{diag}(\mathbf{Z}^v) = 0 \tag{3}$$

where α is a trade-off parameter, and $|\cdot|_F^2$ is the Frobenius norm. For each spatial scale, the first term $|\mathbf{Y}^l - \mathbf{Y}^l\mathbf{Z}^l|_F^2$ quantifies the approximation error in the sparse representation by other feature vectors, and the term $\|\mathbf{Z}^l\|_F^2$ guarantees the sparse and block diagonal solution of $\mathbf{Z}^l$.

Meanwhile, each graph matrix $\mathbf{Z}^l$ has different contributions to the fused graph $\mathbf{S}$, since some spectral features at certain spatial scales may contain irrelevant representations for band selection. Accordingly, MSSC assumes that the graph $\mathbf{Z}^l$ at each scale is a perturbation of the fused graph $\mathbf{S}$, and the $\mathbf{Z}^l$ closer to $\mathbf{S}$ should have a larger weight, which can be expressed as [45]

$$\min_{w^l} \sum_{l=0}^{L} w^l \left\|\mathbf{Z}^l - \mathbf{S}\right\|_F^2 \tag{4}$$

where w^l weights the contribution of the graph at different spatial scales to $\mathbf{S}$.

Using multiview learning, the fused graph $\mathbf{S}$ uncovers the cluster structure of all spectral bands at different spatial scales, i.e., spectral features should be grouped into k different clusters on the fused graph $\mathbf{S}$. Following the Ky Fan's theorem [46], if the constraint $\sum_{i=1}^{k} \lambda_i = 0$ is satisfied, the fused graph $\mathbf{S}$ has an ideal neighbor assignment and the band points are clustered into k clusters [47]. We can obtain an objective function term by minimizing the multiplicity of zero as an eigenvalue of its Laplacian matrix $\mathbf{L}$

$$\sum_{i=1}^{k} \lambda_i = \min_{\mathbf{U}} \operatorname{Tr}(\mathbf{U}^{\mathrm{T}}\mathbf{L}\mathbf{U}), \text{ s.t., } \mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

where $\mathrm{Tr}(\cdot)$ is the trace operator, $\mathbf{I} \in \mathbb{R}^{k\times k}$ is an identity matrix, and $\mathbf{U} \in \mathbb{R}^{N\times k}$ is the k eigenvectors corresponding to the k smallest non-zero eigenvalues of Laplacian matrix $\mathbf{L} = \mathbf{D_S} - (\mathbf{S}+\mathbf{S}^{\mathrm{T}})/2$, $\mathbf{D_S}$ is a diagonal matrix and its elements are column sums of $(\mathbf{S}+\mathbf{S}^{\mathrm{T}})/2$.

By combing (3)–(5), we obtain the MSSC model as

$$\min_{\mathbf{Z}^l,\mathbf{S},\mathbf{U},w_l} \sum_{l=0}^{L} \left\{ \begin{array}{c} \left\|\mathbf{Y}^l - \mathbf{Y}^l\mathbf{Z}^l\right\|_F^2 + \alpha\left\|\mathbf{Z}^l\right\|_F^2 \\ +\beta w^l \left\|\mathbf{Z}^l - \mathbf{S}\right\|_F^2 \end{array} \right\} + \gamma\,\mathrm{Tr}\left(\mathbf{U}^{\mathrm{T}}\mathbf{L}\mathbf{U}\right) \quad \text{s.t., } \mathbf{Z}^l \geq 0, \mathrm{diag}\left(\mathbf{Z}^l\right) = 0, \text{ and } \mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I} \quad (6)$$

where α, β, and γ are three regularization parameters. The third term in (6) correlates closely with spectral clustering, and the weight of each spectral graph $\mathbf{Z}^l$ can be automatically tuned. Therefore, the above model can combine the complementary information of all spectral features at different spatial scales, and guarantee the high-quality clustering result on the fused graph.

We implement the iterative update scheme [48] to optimize (4). At the tth iteration, when fixing $\mathbf{S}_{(t)}$ and $w^l_{(t)}$, $\mathbf{Z}^l_{(t+1)}$ can be optimized by solving the following problem:

$$\min_{\mathbf{Z}^l} \sum_{l=0}^{L} \left\{ \begin{array}{c} \left\|\mathbf{Y}^l - \mathbf{Y}^l\mathbf{Z}^l\right\|_F^2 + \alpha\left\|\mathbf{Z}^l\right\|_F^2 + \\ \beta w^l_{(t)} \left\|\mathbf{Z}^l - \mathbf{S}_{(t)}\right\|_F^2 \end{array} \right\}. \quad \text{s.t., } \mathbf{Z}^l \geq 0, \mathrm{diag}\left(\mathbf{Z}^l\right) = 0 \quad (7)$$

Taking the derivation with regard to $\mathbf{Z}^l$ and setting to zero, the update equation for $\mathbf{Z}^l_{(t+1)}$ can be found as $\mathbf{Z}^l_{(t+1)} = ((\mathbf{Y}^l)^{\mathrm{T}}\mathbf{Y}^l + \alpha\mathbf{I} + \beta w^l_{(t)}\mathbf{I})^{-1}(\beta w^l_{(t)}\mathbf{S}_{(t)} + (\mathbf{Y}^l)^{\mathrm{T}}\mathbf{Y}^l)$. When fixing $\mathbf{Z}^l_{(t+1)}$ and $\mathbf{U}_{(t)}$, $\mathbf{S}_{(t+1)}$ can be optimized by solving

$$\min_{\mathbf{S}} \sum_{l=0}^{L} \left\|\mathbf{Z}^l_{(t)} - \mathbf{S}\right\|_F^2 + \gamma\,\mathrm{Tr}\left(\mathbf{U}^{\mathrm{T}}_{(t)}\mathbf{L}\mathbf{U}_{(t)}\right). \quad (8)$$

The fused graph matrix $\mathbf{S}_{(t+1)}$ can be updated in columns as $\mathbf{S}_{(t+1)}(:,i) = \sum_l w^l_{(t)}\mathbf{Z}^l_{(t+1)}(:,i) - (r\rho_i/4\beta)/\sum_{\mathbf{l}} w^l_{(t)}$, where $\rho_i \in \mathbb{R}^{MN\times 1}$ is a column vector with $\rho_{ij} = \|\mathbf{U}_{i,:} - \mathbf{U}_{j,:}\|_F^2$. When fixing other variables, $\mathbf{U}_{(t+1)}$ can be updated as

$$\min_{\mathbf{U}} \mathrm{Tr}\left(\mathbf{U}^{\mathrm{T}}\mathbf{L}\mathbf{U}\right), \text{ s.t.,} \mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I}. \quad (9)$$

The optimal solution $\mathbf{U}_{(t+1)}$ is estimated as the k eigenvectors of $\mathbf{L}$ corresponding to the k smallest non-zero eigenvalues. By fixing $\mathbf{Z}^l_{(t+1)}$ and $\mathbf{S}_{(t+1)}$, $w^l_{(t+1)}$ can be updated by using the inverse distance weight scheme as

$$w^l_{(t+1)} = \frac{1}{2\left\|\mathbf{Z}^l_{(t+1)} - \mathbf{S}_{(t+1)}\right\|_F}. \quad (10)$$

The above iterations are repeated until the convergence condition $|\mathbf{S}_{(t+1)} - \mathbf{S}_{(t)}|/\mathbf{S}_{(t)} < 10^{-4}$ or the maximum iteration time $t_{\max} > 500$ is satisfied.

C. Selecting Bands Using Spectral Clustering on the Fused Graph

With the estimation of fused graph matrix $\hat{\mathbf{S}} = \mathbf{S}_{(t+1)} \in \mathbb{R}^{D\times D}$, we adopt spectral clustering [49] to cluster all the projected bands on the fused graph into their subspaces. First, we estimate the symmetric normalized Laplacian matrix from the similarity matrix using $\mathbf{L}_{\mathrm{sym}} = \mathbf{F}^{-1/2}\hat{\mathbf{S}}\mathbf{F}^{-1/2}$,where $\mathbf{F}$ is a diagonal matrix constructed with diagonal elements of $\hat{\mathbf{S}}$. The k eigenvectors $\mathbf{U}_k = [u_1 u_2, \ldots, u_k] \in \mathbb{R}^{D\times k}$ corresponding to the k smallest non-zero eigenvalues are found using the SVD composition of the matrix $\mathbf{L}_{\mathrm{sym}}$. Each row vector $\mathbf{h}_i$ of $\mathbf{U}_k$ is normalized into unit norm using $h_{ij} = u_{ij}/(\sum_k u^2_{ik})^{1/2}$. Finally, we cluster row vectors in the normalized $\mathbf{U}_k$ into k clusters. The bands with row vectors that are closest to its cluster centroid in terms of Euclidean distance formulate the final band subset. The procedure of band selection using MSFGF is summarized in Algorithm 1.

III. Experimental Results

A. Three HSI Datasets

The first data is the Indian Pines dataset, which was taken from the Multispectral Image Data Analysis System group at Purdue University (https://engineering.purdue.edu/∼biehl/MultiSpec/aviris_documentation.html). It was collected by the AVIRIS sensor from JPL on June 12, 1992, covering an area of 6 miles west of West Lafayette, Indiana. The data has 20 m spatial resolutions and 10 nm spectral resolutions ranging from 200 to 2400 nm. The implemented small image has the size of 145 × 145 pixels, and has 200 bands after some necessary image processing works like radiometric corrections and bad band removal. Fig. 2 illustrates 16 classes of main ground objects in the image scene.

The second data is the Pavia University (PaviaU) dataset, which was downloaded from the Computational Intelligence Group in the Basque University (http://www.ehu.es/ccwintco / index.php/Hyperspectral_Remote_Sensing_Scenes). It was acquired by the ROSIS sensor, covering the area of Pavia University. The original images have 1.3 m spatial resolutions and 115 bands. The implemented small data uses 103 bands after removing noisy bands, and has the image size of 610 × 340 pixels. The information of nine classes of main ground objects is shown in Fig. 3.

The third data is the Salinas dataset, which was also obtained from the Computational Intelligence Group in the Basque University (http://www.ehu.eus/ccwintco/index.php/Hypersp- ectral_Remote_Sensing_Scenes). It was acquired by the AVIRIS sensor over the Salinas Valley, California, USA. The original data has 3.7 m spatial resolutions and 224 bands. After the preprocessing works with radiometric corrections and removing bad bands, a small image with 512 × 217 pixels and 204 bands were used in the experiments. Fig. 4 shows the 16 classes of main ground objects in the image scene.

B. Experimental Results

We design four experiments to investigate the performance of MSFGF in hyperspectral classification. The first experiment explores the role of MSLRD and MSSC in the proposed MSFGF for band selection. The second experiment compares the behaviors of MSFGF with state-of-the-art methods under the condition of different numbers of selected bands. The third one compares the performance of MSFGF with other methods in classification when changing the proportions of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Procedure of MSFGF

Input: the HSI data cube $\mathcal{X} = [\mathbf{X}_i]_{i=1}^{D} \in \mathbb{R}^{M\times N\times D}$, the band subset size k, $L = round(\log_2(\min(M, N))) + 1$, parameters $\alpha > 0$, $\beta > 0$, and $\gamma > 0$

Step1: *Constructing the MSLRD model on the HSI data cube in (1)*

Step2: *Extracting multiscale spectral features by optimizing the program (2) using ADMM in [40]*

Step3: *Formulating the MSCC model on the multiscale spectral features in (6)*

Step4: *Constructing the fused graph by solving (6) using the iterative update scheme*

Initialization: $\mathbf{S}_{(0)} = random(D, D), \mathbf{U}_{(0)} = random(D, k), w_{(0)}^{l} = 1/L$

While $(|\mathbf{S}_{(t+1)} - \mathbf{S}_{(t)}|/\mathbf{S}_{(t)} \geq 10^{-4})$ *or* $(t \leq 500)$ **do**

1) update the variable $\mathbf{Z}_{(t+1)}^{l}$ as $\mathbf{Z}_{(t+1)}^{l} = ((\mathbf{Y}^{l})^{\mathrm{T}}\mathbf{Y}^{l} + \alpha\mathbf{I} + \beta w_{(t)}^{l}\mathbf{I})^{-1}(\beta w_{(t)}^{l}\mathbf{S}_{(t)} + (\mathbf{Y}^{l})^{\mathrm{T}}\mathbf{Y}^{l})$;

2) Update $\mathbf{S}_{(t+1)}$ in columns as $\mathbf{S}(:, i)_{(t+1)} = \sum_l w_{(t)}^{l}\mathbf{Z}_{(t+1)}^{l}(:, i) - \frac{r\rho_i}{4\beta}/\sum_{\mathbf{l}} w_{(t)}^{l}$;

3) update $\mathbf{U}_{(t+1)}$ as the k eigenvectors of $\mathbf{L}$ corresponding to the k smallest eigenvalues, where $\mathbf{L} = \mathbf{D_S} - \mathbf{S}_{(t+1)} + \mathbf{S}_{(t+1)}^{\mathrm{T}}/2$, $\mathbf{D_S}$ is a diagonal matrix with elements are column sums of $\mathbf{S}_{(t+1)} + \mathbf{S}_{(t+1)}^{\mathrm{T}}/2$;

4) Update $w_{(t+1)}^{l}$ as $w_{(t+1)}^{l} = 1/2\|\mathbf{Z}_{(t+1)}^{l} - \mathbf{S}_{(t+1)}\|_F$ in (10)

$t \leftarrow t + 1$;

End While

Output: The fused graph matrix $\hat{\mathbf{S}} = \mathbf{S}_{(t+1)}$;

Step5: *Selecting bands using spectral clustering on the fused graph*

1) Construct the similarity matrix as $\hat{\mathbf{S}} + \hat{\mathbf{S}}^{\mathrm{T}}/2$;

2) Group the similarity matrix into k clusters using spectral clustering;

3) Select band indices that normalized row vectors in $\mathbf{U}_k$ are closest to its cluster centroid;

Output: the band subset $\mathbf{M} = \mathcal{X}(:, :, \theta)$.



Fig. 2. Indian Pines image.



Fig. 3. PaviaU image.

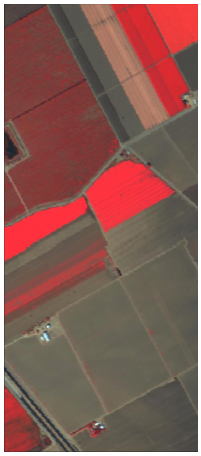


Fig. 4. Salinas image.

training samples. The fourth experiment analyzes parameter settings of three key parameters in MSFGF. The fifth experiment investigates the computational speeds of MSFGF. In the experiments, the support vector machine (SVM) and random forest (RF) are implemented on the selected bands. The radial basis functions are used as the kernel function, and two key parameters including the variance parameter and penalization factor are estimated via cross-validation [50]. We manually set the number of trees in RF as 500, and use the overall classification accuracy (OCA) to quantify classification accuracies.

1) Contributions of MSLRD and MSSC in MSFGF: We compare our MSFGF with "MSLRD + MSC" and "Random + MSSC" to investigate the contributions of MSLRD and MSSC in the proposed method for classification. The difference between MSFGF and "MSLRD + MSC" is that the latter implements the regular multiview spectral clustering (MSC) [51], and directly averages all the individual similarity matrices from multiscale spectral features to obtain the final fused graph. The difference between MSFGF and "Random + MSSC" is that the latter implements the $L+1$ randomly-selected spectral band subsets to construct the multiview spectral graphs for the MSSC. In detail, we randomly select 50 spectral bands from the original set with $L+1$ times to obtain multiview spectral features. For the Indian Pines, the parameters α, β, and γ in MSFGF are manually set to be 1, 10, and 0.001; the α, β, and γ on PaviaU are set to be 100, 1000, and 10^{-3}; the α, β, and γ in MSFGF on Salinas are set to be 10, 1000, and 0.005. The MSSC has

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

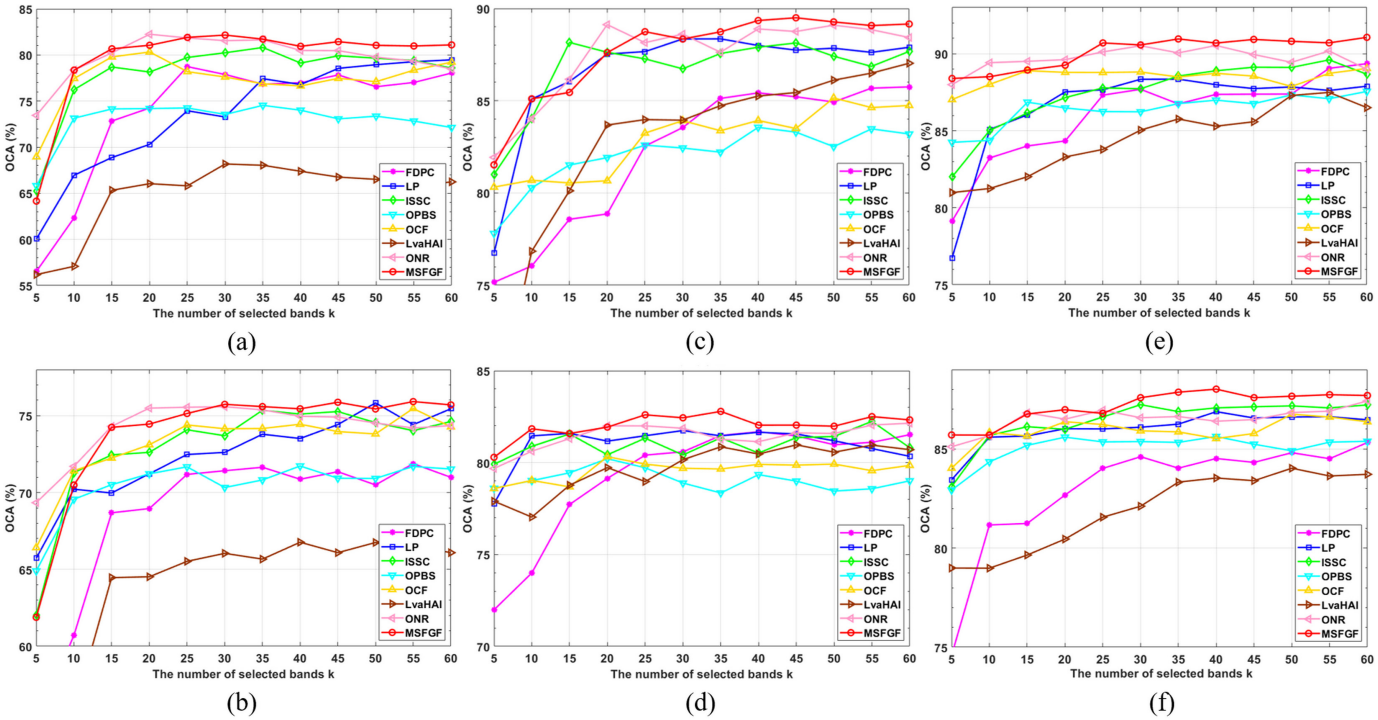


Fig. 5. OCA curves of MSFGF and other seven state-of-the-art band selection methods from SVM and RF classifiers. SVM: (a) Indian Pines (10% of training), (c) PaviaU (1% of training), and (e) Salinas (1% of training). RF: (b) Indian Pines (10% of training), (d) PaviaU (1% of training), and (f) Salinas (1% of training).

TABLE II
COMPARISON IN OCA BETWEEN MSFGF AND TWO OTHER COMBINED SCHEMES

Datasets	Methods	OCA (%)				
		k=10	k=20	k=30	k=40	k=50
Indian Pines (10% of training)	MSLRD+MSC	76.36	78.70	80.61	81.05	80.87
	MSFGF	78.35	81.04	82.15	81.93	81.54
	Random+MSSC	75.22	76.88	77.72	79.34	79.77
PaviaU (1% of training)	MSLRD+MSC	83.41	85.85	86.19	87.58	87.82
	MSFGF	85.10	87.61	88.34	89.35	89.27
	Random+MSSC	83.03	84.26	86.20	86.67	86.87
Salinas (1% of training)	MSLRD+MSC	88.12	89.12	89.42	89.82	89.98
	MSFGF	88.51	89.26	90.57	90.70	90.81
	Random+MSSC	87.94	88.70	88.90	89.09	89.50

the same parameter settings as those of MSFGF on the three datasets. In the experiments, 10% of labeled samples from each class on the Indian Pines, and 1% of labeled samples on PaviaU and Salinas are randomly selected for training, and the rest are used for testing. We run each experiment ten times, and the OCA is derived from the average of ten independent experiments.

Table II lists the comparison results between MSFGF with those of "MSLRD + MSC" and "Random + MSSC," where the number of selected bands k changes from 10 to 50 with a step interval of 10. The MSFGF obtains higher OCAs than "MSLRD +MSC" and "Random + MSSC" on the three datasets. Particularly, the MSFGF promotes up to 3.5% OCAs of "Random + MSSC" on the Indian Pines, and improves up to 2.2% OCA of "MSLRD + MSC" on the PaviaU. That tells both MSLRD and MSSC positively impacts the behaviors of MSFGF in selecting proper bands. More interestingly, the "MSLRD + MSC" performs better than "Random + MSSC" in OCAs, which shows multiscale spectral features from MSLRD have more significant contributions in MSFGF than the fusion procedure by MSSC.

2) Classification Behaviors With Different Number of Selected Bands: Seven state-of-the-art methods are implemented, including a ranking-based method FDPC [14], three searching-based method linear prediction (LP) [52], orthogonal projections based band selection (OPBS) [53] and optimal neighborhood reconstruction (ONR) [54], a sparsity-based graph learning method ISSC, a clustering-based method optimum clustering framework (OCF) [55], and a graph learning-based method local-view-assisted discriminative band selection method with hypergraph autolearning (LvaHAl) [56]. All the methods except LP are newly proposed band selection methods. For each method, we change the k from 5 to 60 with a step interval of 5. For the LvaHAI, using cross-validation, the parameters λ_1, β_1, and r_1 on the Indian Pines are set to be 0.1, 10^{-7}, and 2.5, respectively, the λ_1, β_1, and r_1 on the Indian Pines are set to be 0.05, 10^{-6}, and 2.0, respectively, and the λ_1, β_1, and r_1 on the Salinas are set to be 0.1, 10^{-7}, and 2.0, respectively. The three parameters α, β, and γ of MSFGF on three datasets are the same with those in experiment Section III-B1.

Fig. 5 shows the OCA curves of MSFGF and the other seven methods on the three datasets. The LP, ISSC, ONR curves are higher than LvaHAI, OCF, and OPBS on the three datasets, but they are inferior to MSFGF with k larger than

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

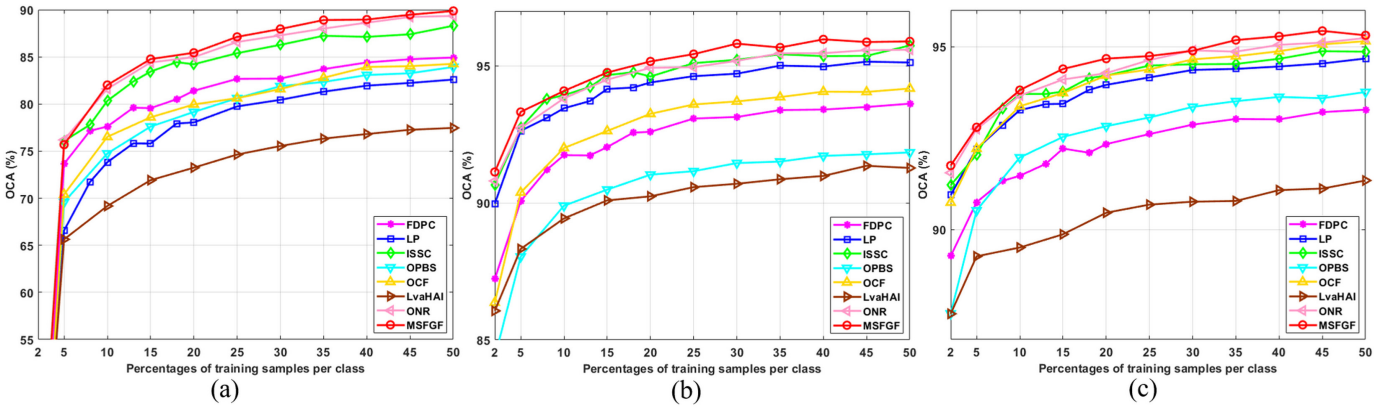


Fig. 6. OCA curves of MSFGF and other seven state-of-the-art band selection methods from SVM (a) Indian Pines, (b) PaviaU, and (c) Salinas.

TABLE III
CLASSIFICATION ACCURACIES OF ALL THE METHODS ON THREE DATASETS IN THE CASE OF 40, 50, 60 BANDS

Datasets	Classifier	bands	OCA(%)							
			LP	FDPC	ISSC	OPBS	OCF	LvaHAI	ONR	MSFGF
Indian Pines (10% of training)	**SVM**	k=40	76.78±0.99	76.98±1.58	79.12±1.27	74.00±0.45	76.63±0.67	67.38±0.73	80.48±0.55	80.93±0.44
		k=50	78.94±0.41	76.55±0.70	79.64±1.14	73.35±0.67	77.10±0.97	66.50±0.57	79.76±0.46	81.04±0.37
		k=60	79.47±0.84	78.03±0.77	78.57±0.76	72.12±0.21	79.21±0.26	66.23±0.49	78.47±0.42	81.08±0.22
	RF	k=40	73.51±0.59	70.86±0.42	75.10±0.53	71.72±0.68	74.44±0.43	66.76±0.14	74.96±0.76	75.44±0.31
		k=50	75.83±0.14	70.49±0.47	74.55±0.90	70.91±1.59	73.81±1.00	66.75±0.68	74.52±0.49	75.43±0.28
		k=60	75.45±0.69	70.99±0.88	74.62±0.41	71.52±0.74	74.27±0.63	66.08±0.85	74.35±0.72	75.70±0.64
PaviaU (1% of training)	**SVM**	k=40	87.99±0.52	85.42±0.98	87.90±1.42	83.55±0.87	83.93±1.33	85.28±0.72	88.89±1.16	89.35±0.84
		k=50	87.85±0.55	84.93±0.84	87.41±0.31	82.51±1.11	85.15±1.01	86.11±0.14	89.09±0.46	89.27±0.54
		k=60	87.90±1.12	85.75±2.52	87.67±0.62	83.20±0.39	84.74±2.51	87.03±0.69	88.43±0.68	89.16±0.32
	RF	k=40	81.65±0.61	81.68±0.24	80.52±0.71	79.34±0.51	79.90±0.19	80.46±0.63	81.14±1.00	82.04±0.41
		k=50	81.19±0.66	80.97±0.10	81.42±0.81	78.45±1.07	79.92±1.04	80.56±0.85	81.58±0.64	81.98±0.50
		k=60	80.34±0.29	81.51±0.77	80.81±0.27	79.01±0.17	79.85±0.74	80.70±0.75	82.14±0.90	82.32±0.65
Salinas (1% of training)	**SVM**	k=40	88.43±1.16	87.36±0.69	88.70±0.64	86.99±0.26	88.73±1.04	85.30±3.11	90.54±0.51	90.70±0.36
		k=50	88.17±2.55	87.39±0.51	89.01±1.01	87.33±1.01	87.88±2.00	87.30±1.27	89.45±1.55	90.81±0.43
		k=60	88.21±0.12	89.36±0.64	88.57±0.89	87.56±2.12	89.05±0.52	86.51±0.60	88.99±0.83	91.07±0.10
	RF	k=40	86.88±0.44	84.52±0.31	87.07±0.68	85.63±0.10	85.52±0.24	83.53±1.10	86.40±0.39	88.01±0.09
		k=50	86.61±0.89	84.81±0.50	87.17±0.14	84.90±0.89	86.75±0.26	84.02±0.60	86.83±0.19	87.66±0.10
		k=60	86.47±0.98	85.32±0.66	87.22±0.62	85.38±0.49	86.37±0.49	83.72±1.44	87.42±0.56	87.69±0.58

TABLE IV
STATISTICAL SIGNIFICANCE OF THE MCNEMAR'S TEST ON INDIAN PINES IN THE CASE OF 40, 50, 60 BANDS

Number of bands k	LP		FDPC		ISSC		OPBS		OCF		ONR		LvaHAI	
	SVM	RF	SVM	RF	SVM	RF	SVM	RF	SVM	RF	SVM	RF	SVM	RF
40	-5.45	-4.41	-5.46	-10.81	-3.49	-1.48	-10.46	-7.69	-5.44	-2.08	-0.43	-2.29	-19.22	-17.27
50	-4.07	0.62	-6.14	-11.17	-3.03	-2.60	-11.32	-8.91	-4.29	-4.31	-2.79	-2.79	-18.85	-17.68
60	-2.27	-1.44	-5.14	-10.23	-4.14	-2.78	-12.95	-8.61	-3.79	-3.14	-4.28	-2.87	-19.28	-18.30

30 on the three datasets. The MSFGF curves from SVM are higher than those of ONR, and particularly it clearly outperforms ISSC on all three datasets. The reason for that is MSFGF not only considers various clustering structures of spectral features at different spatial scales, but also fuses them to be a consensus graph better than that of ISSC. More interestingly, the graph learning-based method LvaHAI offers divergent performance across the three HSI datasets. It has the lowest curves on the Indian Pines and Salinas, but performs better than OPBS and OCF on the PaviaU with a greater k. The reason may be that the performance of LvaHAI is highly correlated with the dimensionality of original bands and the number of training samples [56]. The conflict between the small number of training samples (e.g., 10% of Indian Pines and 1% of Salinas) and the high-dimensionality of 200 bands cannot be accommodated by the LvaHAI, therefore yielding poor classification accuracy. Table III shows the OCAs and standard deviations (STDs) of different methods on three datasets in the case of 40, 50, 60 bands. It can be seen that the proposed MSFGF method provides the best overall classification results on two classifiers over three datasets.

In order to further investigate whether the accuracy of our method is statistically significant with the other seven methods, we use the McNemar's test to evaluate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

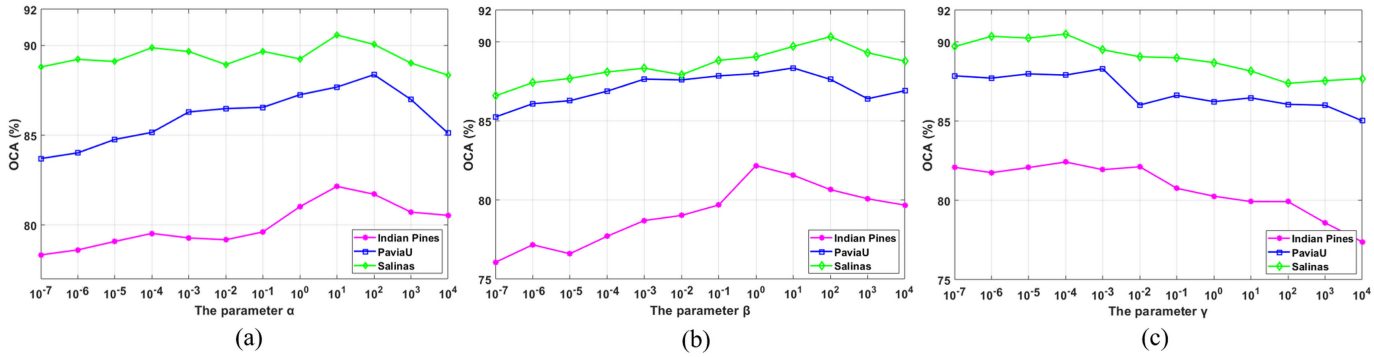


Fig. 7. OCA curves of MSFGF by changing three parameters (a) α, (b) β, and (c) γ.

TABLE V
COMPUTING TIME OF EIGHT BAND SELECTION METHODS WITH DIFFERENT SIZES OF BAND SUBSET k

Datasets	Number of selected Bands k	Computational times (Seconds)							
		FDPC	LP	ISSC	OPBS	OCF	LvaHAI	ONR	MSFGF
Indian Pines	k=10	1.570	19.983	0.334	0.102	0.413	64.819	0.504	28.091
	k=20	1.613	28.832	0.536	0.147	0.592	64.823	0.522	28.432
	k=30	1.794	39.095	0.687	0.188	0.789	64.833	0.537	28.808
	k=40	1.870	58.159	0.902	0.212	0.957	64.840	0.546	29.237
	k=50	1.980	82.683	2.018	0.230	1.141	64.850	0.572	29.671
PaviaU	k=10	6.180	107.434	0.759	0.302	1.068	472.220	0.269	140.292
	k=20	6.299	153.329	0.774	0.314	1.117	472.237	0.271	140.608
	k=30	6.359	221.934	0.985	0.322	1.207	472.340	0.273	140.689
	k=40	6.650	260.686	0.997	0.330	1.309	472.458	0.301	140.716
	k=50	6.682	336.872	1.178	0.341	1.412	472.571	0.312	140.727
Salinas	k=10	7.291	83.470	0.880	0.426	1.233	1591.926	0.620	144.978
	k=20	7.415	117.454	0.904	0.431	1.437	1591.933	0.639	145.105
	k=30	7.449	179.980	1.206	0.445	1.617	1591.939	0.645	145.293
	k=40	7.652	261.064	1.330	0.453	1.797	1591.945	0.653	145.539
	k=50	7.711	364.909	1.690	0.462	2.064	1592.172	0.670	145.867

different approaches [57]. Table IV shows the test results of Z-statistics computed by McNemar's test when comparing MSFGF to the other seven methods on Indian Pines dataset. The level of significance is 5%. If $|Z| > 1.96$, the difference between the method is statistically significant. Negative values ($Z < -1.96$) indicate that MSFGF has better accuracy. Conversely, other methods are better. As shown in Table IV, MSFGF is significantly better than ISSC in the case of RF with 40 bands, LP in the case of RF with 60 bands, and ONR in the case of SVM with 40 bands. In addition, the difference between LP and MSFGF is not significant in the case of 50 bands. The rest of the Z values are negative and their absolute values are greater than 1.96. Thus, MSFGF has a statistically significant difference from other seven methods.

3) Classification Performance With Different Proportions of Training Samples: We compare the OCAs of MSFGF with those of the other seven methods, by changing the percentage of training samples per class from 2 to 50. The selected bands of all involved methods are manually set to be 30. Fig. 6 shows the OCA curves of all eight methods from the SVM classifier. All the curves rise with the increasing number of training samples. The LvaHAI curves are the lowest. The OPBS outperforms the LvaHAI, but their curves are lower than those of the OCF. The ISSC obtains higher OCAs than LP, OPBS, FDPC, and LvaHAI, but its three curves are slightly lower than those of the ONR and MSFGF. The MSFGF performs slightly better than the ONR and achieves the highest OCA curves. The advantage of MSFGF against other methods further supports the observations in Section III-B2.

4) Performance Sensitivity of MSFGF to Three Key Parameters: We change the three key parameters α, β, and γ to explore their impacts on the classification performance of MSFGF. The selected bands are set to be 30, and the proportions of training samples per class on Indian Pines, PaviaU, and Salinas are set to be 10%, 1%, and 1%, respectively. We manually set the other two parameters, and change the third one within the interval [10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10, 100, 10^3, 10^4]. For the experiment of parameter α, we set β and γ on Indian Pines as 10 and 0.001, the PaviaU as 10^3 and 10^{-3}, and 10^3 and 10^{-4}, respectively. For the experiments of parameter β, we set α and γ on Indian Pines as 1 and 10^{-2}, the PaviaU as 10^3 and 10^{-3}, and 10 and 10^{-3}, respectively. For the experiments of parameter γ, we set α and β on Indian Pines as 0.1 and 10, the PaviaU as 100 and 10^3, and 10 and 10^2, respectively.

As shown in Fig. 7(a), the OCA curves of MSFGF gradually rise with the increasing α, achieve the highest when α is between 10 and 100, and then begin to decrease. The similar phenomena exist in Fig. 7(b) of the parameter β, and the OCA achieves the highest when β is between 1 and 100. The OCAs in Fig. 7(c) is relatively stable when γ changes from 10^{-7} to 10^{-4}, and begins to decrease until the end

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

of the interval. Therefore, a moderate α and β between 10 and 100 are recommended, and the γ smaller than 10^{-4} is recommended to bring about good classification result of MSFGF.

5) Computational Time Against Other Methods: We further compare the computational time of MSFGF with the above seven methods, by changing the number of selected bands from 10 to 50 with a step interval of 10. All the eight methods are implemented in MATLAB 2020a and their codes are run on a WIN10 computer with AMD Ryzen 9 3950X 16-Core Processor and 128 GB of RAM.

As shown in Table V, the OPBS spends the least computational time, and followed by the ISSC, OCF, and ONR. The ISSC is more time-consuming than OPBS and ONR, but its computational speed surpasses other five methods. The FDPC has slower computational speed than OPBS, ISSC, and ONR, but runs faster than LP, LvaHAI, and MSFGF. The MSFGF takes shorter computational time than LP and LvaHAI, and the LvaHAI is the slowest method that requires up to 27 min on the Salinas data. The descending order of computational speeds of all the methods is: OPBS, ONR, ISSC, OCF, FDPC, MSFGF, LP, LvaHAI. Therefore, our MSFGF has moderate computational performance among all methods.

IV. Conclusion

This article presents a MSFGF method to explore the spatial-scale issue of various ground objects when selecting proper bands from hyperspectral images. The MSFGF adopts the MSLRD to extract low-rank spectral features of ground objects at various spatial scales, utilizes the MSSC to combine the complementary information of multiscale spectral features to formulate a consensus graph, and uses spectral clustering to select bands on the fused graph. Five experiments on three hyperspectral datasets are designed to validate the performance of MSFGF. Experimental results show that the MSFGF performs slightly better than the ONR and obviously better than graph learning methods ISSC and LvaHAI and other four state-of-the-art methods, with a relatively lower computational cost. The MSLRD and MSSC positively contribute to the good performance of MSFGF in selecting bands for classification. In realistic applications, parameters α and β are recommended to be set between 10 and 100 and parameter γ as smaller than 10^{-4}.

References

[1] Y. Zhong *et al.*, "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[2] R. Pu, *Hyperspectral Remote Sensing: Fundamentals and Practices*. Boca Raton, FL, USA: CRC Press, 2017.

[3] C. Camino, V. González-Dugo, P. Hernández, J. C. Sillero, and P. J. Zarco-Tejada, "Improved nitrogen retrievals with airborne-derived fluorescence and plant traits quantified from VNIR-SWIR hyperspectral imagery in the context of precision agriculture," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 70, pp. 105–117, Aug. 2018.

[4] L. Liu, N. C. Coops, N. W. Aven, and Y. Pang, "Mapping urban tree species using integrated airborne hyperspectral and LiDAR remote sensing data," *Remote Sens. Environ.*, vol. 200, pp. 170–182, Oct. 2017.

[5] Y. Tan, L. Lu, L. Bruzzone, R. Guan, Z. Chang, and C. Yang, "Hyperspectral band selection for lithologic discrimination and geological mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 471–486, Jan. 2020.

[6] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[7] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[8] J. P. Rivera-Caicedo, J. Verrelst, J. Muñoz-Marí, G. Camps-Valls, and J. Moreno, "Hyperspectral dimensionality reduction for biophysical variable statistical retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 132, pp. 88–101, Oct. 2017.

[9] F. Luo, L. Zhang, B. Du, and L. Zhang, "Dimensionality reduction with enhanced hybrid-graph discriminant learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5336–5353, Aug. 2020.

[10] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[11] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.

[12] W. Sun and Q. Du, "Hyperspectral band selection a review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[13] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[14] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2015.

[15] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5028–5039, Jun. 2020.

[16] S. Sawant and P. Manoharan, "Hyperspectral band selection based on Metaheuristic optimization approach," *Infr. Phys. Technol.*, vol. 107, Jun. 2020, Art. no. 103295.

[17] J. Tschannerl *et al.*, "MIMR-DGSA: Unsupervised hyperspectral band selection based on information theory and a modified discrete gravitational search algorithm," *Inf. Fusion*, vol. 51, pp. 189–200, Nov. 2019.

[18] R. Zhang and J. Ma, "Feature selection for hyperspectral data based on recursive support vector machines," *Int. J. Remote Sens.*, vol. 30, no. 14, pp. 3669–3677, 2009.

[19] W. Sun, G. Yang, J. Peng, and Q. Du, "Hyperspectral band selection using weighted Kernel regularization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3665–3676, Sep. 2019.

[20] J. Feng *et al.*, "Convolutional neural network based on bandwise-independent convolution and hard thresholding for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 251–267, Jun. 2020.

[21] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Nov. 2020.

[22] W. Sun, W. Li, J. Li, and Y. M. Lai, "Band selection using sparse nonnegative matrix factorization with the thresholded Earth's mover distance for hyperspectral imagery classification," *Earth Sci. Informat.*, vol. 8, no. 4, pp. 907–918, Dec. 2015.

[23] W. Sun, L. Tian, Y. Xu, D. Zhang, and Q. Du, "Fast and robust self-representation method for hyperspectral band selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 5087–5098, Nov. 2017.

[24] Y. Feng, Y. Yuan, and X. Lu, "A non-negative low-rank representation for hyperspectral band selection," *Int. J. Remote Sens.*, vol. 37, no. 19, pp. 4590–4609, Oct. 2016.

[25] J. Yin, Y. Wang, and Z. Zhao, "Optimal band selection for hyperspectral image classification based on inter-class separability," in *Proc. Symp. Photon. Optoelectron.*, Jun. 2010, pp. 1–4.

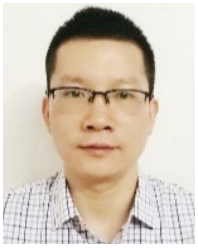
[26] A. Datta, S. Ghosh, and A. Ghosh, "Combination of clustering and ranking techniques for unsupervised band selection of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2814–2823, Jun. 2015.

[27] S. Li, H. Wu, D. Wan, and J. Zhu, "An effective feature selection method for hyperspectral image classification based on genetic algorithm and support vector machine," *Knowl.-Based Syst.*, vol. 24, no. 1, pp. 40–48, Feb. 2011.

[28] Y. Jiang and C. Li, "mRMR-based feature selection for classification of cotton foreign matter using hyperspectral imaging," *Comput. Electron. Agricult.*, vol. 119, pp. 191–200, Nov. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.

[29] J. Peng *et al.*, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 99, no. 1, pp. 2–35, Jun. 2021.
[30] L. Zhang and F. Luo, "Review on graph learning for dimensionality reduction of hyperspectral image," *Geo-Spatial Inf. Sci.*, vol. 23, no. 1, pp. 98–106, Jan. 2020.
[31] R. Wang, F. Nie, and W. Yu, "Fast spectral clustering with anchor graph for large hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2003–2007, Nov. 2017.
[32] C. Yu, K. Cen, C. Chang, and F. Li, "Unsupervised hyperspectral band selection method based on low-rank representation," in *Proc. Int. Conf. Commun.*, 2018, pp. 1053–1061.
[33] W. Sun, L. Zhang, B. Du, W. Li, and Y. M. Lai, "Band selection using improved sparse subspace clustering for hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2784–2797, Jun. 2015.
[34] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Squaring weighted low-rank subspace clustering for hyperspectral image band selection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2434–2437.
[35] S. L. Al-khafaji, J. Zhou, A. Zia, and A. W.-C. Liew, "Spectral-spatial scale invariant feature transform for hyperspectral images," *IEEE Trans. Image Process.*, vol. 27, no. 2, pp. 837–850, Feb. 2017.
[36] X. Zhao *et al.*, "Estimation of soybean breeding yield based on optimization of spatial scale of UAV hyperspectral image," *Trans. Chin. Soc. Agricult. Eng.*, vol. 33, no. 1, pp. 110–116, 2017.
[37] W. Sun, W. Shao, J. Peng, G. Yang, X. Meng, and Q. Du, "Multiscale low-rank spatial features for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, early access, Nov. 9, 2021, doi: 10.1109/LGRS.2020.3034631.
[38] B. Recht, M. Fazel, and P. A. Parrilo, "Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization," *SIAM Rev.*, vol. 52, no. 3, pp. 471–501, 2010.
[39] F. Ong and M. Lustig, "Beyond low rank + sparse: Multiscale low rank matrix decomposition," *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 4, pp. 672–687, Jun. 2016.
[40] W. Sun, G. Yang, B. Du, L. Zhang, and L. Zhang, "A sparse and low-rank near-isometric linear embedding method for feature extraction in hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 4032–4046, Jul. 2017.
[41] A. S. Bandeira and R. van Handel, "Sharp nonasymptotic bounds on the norm of random matrices with independent entries," *Ann. Probab.*, vol. 44, no. 4, pp. 2479–2506, 2016.
[42] S. Boyd, N. Parikh, and E. Chu, *Distributed Optimization and Statistical Learning Via the Alternating Direction Method of Multipliers*. Hanover, MS, USA: Now, 2011.
[43] H. Gao, F. Nie, X. Li, and H. Huang, "Multi-view subspace clustering," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 4238–4246.
[44] Z. Kang *et al.*, "Multi-graph fusion for multi-view spectral clustering," *Knowl.-Based Syst.*, vol. 189, Feb. 2020, Art. no. 105102.
[45] F. Nie, J. Li, and X. Li, "Parameter-free auto-weighted multiple graph learning: A framework for multiview clustering and semi-supervised classification," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, 2016, pp. 1881–1887.
[46] K. Fan, "On a theorem of Weyl concerning eigenvalues of linear transformations I," *Proc. Nat. Acad. Sci. USA*, vol. 35, no. 11, p. 652, 1949.
[47] B. Mohar, "The Laplacian spectrum of graphs," in *Graph Theory, Combinatorics, and Applications*, vol. 2. Kalamazoo, MI, USA: Wiley, 1991, pp. 871–898.
[48] L. Marin, L. Elliott, P. Heggs, D. Ingham, D. Lesnic, and X. Wen, "An alternating iterative algorithm for the Cauchy problem associated to the Helmholtz equation," *Comput. Methods Appl. Mech. Eng.*, vol. 192, nos. 5–6, pp. 709–722, 2003.
[49] U. von Luxburg, "A tutorial on spectral clustering," *Statist. Comput.*, vol. 17, no. 4, pp. 395–416, 2007.
[50] L. Xie, G. Li, M. Xiao, L. Peng, and Q. Chen, "Hyperspectral image classification using discrete space model and support vector machines," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 374–378, Mar. 2017.
[51] X. Cao, C. Zhang, H. Fu, S. Liu, and H. Zhang, "Diversity-induced multi-view subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 586–594.
[52] Q. Du and H. Yang, "Similarity-based unsupervised band selection for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 564–568, Oct. 2008.
[53] W. Zhang, X. Li, Y. Dou, and L. Zhao, "A geometry-based band selection approach for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4318–4333, Aug. 2018.
[54] Q. Wang, F. Zhang, and X. Li, "Hyperspectral band selection via optimal neighborhood reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8465–8476, Dec. 2020.
[55] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.
[56] X. Wei, L. Cai, B. Liao, and T. Lu, "Local-view-assisted discriminative band selection with hypergraph autolearning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2042–2055, Mar. 2019.
[57] G. M. Foody, "Thematic map comparison: Evaluating the statistical significance of differences in classification accuracy," *Photogramm. Eng. Remote Sens.*, vol. 70, no. 5, pp. 627–633, May 2004.



Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he studied with the Department of Applied Mathematics, University of Maryland, College Park, MD, USA, where he was a Visiting Scholar with the famous Professor John Benedetto to study on the dimensionality reduction of hyperspectral images. From 2014 to 2016, he studied with the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, where he held a post-doctoral training to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he worked with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA, where he worked as a Visiting Scholar in hyperspectral image processing. He is currently a Full Professor with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo, Zhejiang, China. He has published more than 80 journal articles and his research interest includes hyperspectral image processing with machine learning.



Gang Yang received the M.S. degree in geographical information systems from Hunan University of Science and Technology, Xiangtan, China, in 2012, and the Ph.D. degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2016.

He is currently a Lecturer with Ningbo University, Ningbo, Zhejiang, China. His research interests focus on missing information reconstruction of remote sensing image, cloud removal of remote sensing image, and remote sensing time-series products temporal reconstruction.



Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.



Xiangchao Meng (Member, IEEE) received the B.S. degree in geographic information systems from Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in cartography and geography information system from Wuhan University, Wuhan, China, in 2017.

He is currently a Lecturer with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China. His research interests include remote sensing image fusion and quality evaluation.



Ke He received the B.S. degree in geographic information science from Shandong Jiaotong University, Qingdao, China, in 2020. He is currently pursuing the master's degree with Ningbo University, Ningbo, China.

His research interests are deep learning and hyperspectral imagery.



Wei Li (Senior Member, IEEE) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012.

Subsequently, he spent one year as a Post-Doctoral Researcher with the University of California at Davis, Davis, CA, USA. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. He has authored more than 150 peer-reviewed articles and 100 conference papers which have been cited 7500 times (Google Scholarj. His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li is currently serving as Associate Editor for the IEEE Transactions on Geoscience and Remote Sensing (TGRS), IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), and IEEE Signal Processing Letters (SPL). He was a recipient of the JSTARS Best Reviewer in 2016 and TGRS Best Reviewer Award in 2020 from the IEEE Geoscience and Remote Sensing Society (GRSS), and the Outstanding Paper Award at IEEE International Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (Whispers) in 2019.



Heng-Chao Li (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from Southwest Jiaotong University, Chengdu, China, in 2001 and 2004, respectively, and the Ph.D. degree from the Graduate University of Chinese Academy of Sciences, Beijing, China, in 2008, all in information and communication engineering.

From November 2013 to October 2014, he was a Visiting Scholar working with Prof. W. J. Emery with the University of Colorado Boulder, Boulder, CO, USA. He is a Full Professor with the School of Information Science and Technology, Southwest Jiaotong University. He has authored over 100 research articles, including 65 JCR journal articles, such as published in the *ISPRS J P* and *RS*, the IEEE Transactions on Geoscience and Remote Sensing (TGRS), the IEEE Journal of Selected Topics in Signal Processing (JSTSP), the IEEE Transactions on Image Processing (TIP), and the *Pattern Recognition*. His research interests include statistical analysis of SAR images, remote sensing image processing, and pattern recognition.

Dr. Li is an Editorial Board Member of the *Journal of Southwest Jiaotong University* and the *Journal of Radars*. He is also an Active Reviewer of over 30 international journals. He was a recipient of the 2018 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society for his service to the IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS). Moreover, he has served as a Guest Editor for the Special Issues of the *Journal of Real-Time Image Processing*, the IEEE JSTARS, and the IEEE JMASS, a Program Committee Member for the 26th International Joint Conference on Artificial Intelligence (IJCAI-2017), and the 10th International Workshop on the Analysis of Multitemporal Remote Sensing Images (MULTITEMP-2019), and the Session Chair for the 2017 International Geoscience and Remote Sensing Symposium (IGARSS-2017), and the 2019 Asia-Pacific Conference on Synthetic Aperture Radar (APSAR-2019). He is serving as an Associate Editor for the IEEE JSTARS.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland–Baltimore County, Baltimore, MD, USA, in 2000.

She is currently a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of SPIE–International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She served as a Co-Chair for the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013. She was the Chair with the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She was the General Chair for the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, China, in 2012. She served as an Associate Editor of the IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), the *Journal of Applied Remote Sensing*, and the IEEE Signal Processing Letters. From 2016 to 2020, she was the Editor-in-Chief for the IEEE JSTARS. She is currently a member of the IEEE Periodicals Review and Advisory Committee.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:51:06 UTC from IEEE Xplore. Restrictions apply.